

1. Fundamentals of Probability Theory

Young-geun Kim

Department of Mathematical Sciences

MTH345: Deep Generative Models: A Statistical Perspective



ULSAN NATIONAL INSTITUTE OF
SCIENCE AND TECHNOLOGY

Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- ▶ This lecture briefly reviews preliminary concepts in probability.
- ▶ Useful materials include:
 - ▶ Mathematics for Machine Learning [MML] (Deisenroth et al., 2020)
 - ▶ A First Course in Probability [FCP] (Ross, 2020)
 - ▶ Mathematical Statistics and Data Analysis [MSDA] (Rice, 2007)
- ▶ Selected examples and text excerpts are reproduced from the materials for educational use only under fair use.

Outline

1. PROBABILITY

2. CONDITIONAL PROBABILITY AND EVENT INDEPENDENCE

Probability

- ▶ What is probability? In our daily lives, we frequently use the term *probability* in several different contexts, including:
 1. The limit of relative frequencies
 2. The degree of belief
- ▶ For the first interpretation, consider the random experiment of tossing a coin. Suppose we perform $n = 5$ repeated trials and observe: H, T, H, T, H.
- ▶ We can construct a probability table based on relative frequencies:

Outcome	H	T
Relative Frequency	3/5	2/5

- ▶ Assuming that the coin is fair, as $n \rightarrow \infty$, the sequence of probability tables is expected to converge to:

Outcome	H	T
Limit of Relative Frequency	1/2	1/2

Probability



- ▶ For the second interpretation, someone might say:
"This is a dog with 30% probability."
- ▶ The probability here represents our degree of confidence (0: no confidence, 1: absolute certainty), rather than the limit of relative frequencies.
- ▶ Reflecting this individual's belief, we can construct another probability table:

Outcome	Dog	Not a dog
Degree of Belief	.3	.7

This image was generated by Google Gemini.

Probability



This image was generated by Google Gemini.

Probability

- ▶ Modern probability theory is built upon fundamental assumptions (*axioms*) that unify the notion of probability across these different contexts.
- ▶ We first define the *sample space* S , the set of all possible outcomes. For example,

$$S = \{H, T\}. \quad (1)$$

- ▶ An *event* A is defined as a subset of the sample space. We then introduce the *event space* $\mathcal{A}$, which is the collection of all possible events. For example,

$$\mathcal{A} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}. \quad (2)$$

Here, $\{H\}$ is the event where only heads is observed, and $\{H, T\}$ is the event where either heads or tails is observed (i.e., any possible outcome occurs).

- ▶ Now, the probability $P(\cdot)$ can be viewed as a mapping from $\mathcal{A}$ to $[0, 1]$. For example,

$$P(\emptyset) = 0, \quad P(\{H\}) = 1/2, \quad P(\{T\}) = 1/2, \quad P(\{H, T\}) = 1. \quad (3)$$

Probability

- ▶ Note that both the relative frequency-based and the degree of belief-based interpretations share some fundamental characteristics.
- ▶ **Probability:** We call $P : \mathcal{A} \rightarrow [0, 1]$ a *probability (measure)* if it satisfies:
 1. $P(S) = 1$.
 2. If $A \in \mathcal{A}$, then $P(A) \geq 0$.
 3. If $A_1, A_2, \dots \in \mathcal{A}$ are pairwise disjoint (i.e., $A_i \cap A_j = \emptyset$ for any $i \neq j$), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i). \quad (4)$$

- ▶ In the coin-tossing example,

$$P(\emptyset) = 0, \quad P(\{H\}) = 1/2, \quad P(\{T\}) = 1/2, \quad P(\{H, T\}) = 1. \quad (5)$$

- ▶ In the generated image example,

$$P(\emptyset) = 0, \quad P(\{\text{Dog}\}) = 0.3, \quad P(\{\text{Not a dog}\}) = 0.7, \quad P(\{\text{Dog, Not a dog}\}) = 1. \quad (6)$$

• **Wahl** = election

- | | | | |
|---------|-------|-------|---------|
| Outcome | s_1 | s_2 | $\dots$ |
|---------|-------|-------|---------|

$= 1$. Then, the function F is

(ii) $\nabla \cdot \mathbf{u} = 0$

valid probability measure (i.e. it satisfies the three axioms)

-

Young-geun Kim

Probability

- ▶ When S is uncountably infinite, e.g., $S = [0, 1]$, we cannot construct P in a similar way.
- ▶ Note that because the sum of uncountably many positive numbers diverges, the first axiom $P(S) = 1$ cannot hold. Furthermore, some subsets of S pose many analytic challenges.
- ▶ Instead, we limit our focus to real intervals, (a, b) where $-\infty \leq a \leq b \leq \infty$.
- ▶ We then define $\mathcal{A}$ as the collection of subsets generated by countable set operations (union, intersection, and complement) on these intervals. For instance, when $S = [0, 1]$, we have:

$$(0, 1/4] \cup [3/4, 1) = (0, 1) \cap (1/4, 3/4)^c \in \mathcal{A}. \quad (8)$$

Probability

- ▶ We first define the probability of an interval as:

$$P((a, b)) = \int_a^b f(s) ds \quad (9)$$

for some non-negative function f satisfying $\int_{-\infty}^{\infty} f(s) ds = 1$. Then, the function P extended to countable unions of pairwise disjoint intervals (a_i, b_i) by

$$P\left(\bigcup_{i=1}^{\infty}(a_i, b_i)\right) = \sum_{i=1}^{\infty} P((a_i, b_i)), \quad (10)$$

forms a valid probability measure.

- ▶ For example, when $S = [0, 1]$, if P is defined as

$$P((a, b)) = b - a \left(= \int_a^b 1 \, ds \right), \quad \text{where } 0 \leq a \leq b \leq 1, \quad (11)$$

and is extended to satisfy Equation (10), then P forms a valid probability measure.

Outline

1. PROBABILITY

2. CONDITIONAL PROBABILITY AND EVENT INDEPENDENCE

Conditional Probability

- ▶ Consider the random experiment of tossing a coin twice. The corresponding probability table can be expressed as:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/4	1/4	1/4	1/4

- ▶ Q: Suppose we are informed that the first coin toss resulted in H . Do we still assign the same probability to each outcome?
- ▶ *Conditional probability* provides a principled framework for updating our probabilities based on such newly acquired information.

Conditional Probability

- Without any additional information, the probability table is:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/4	1/4	1/4	1/4

- Given that the first coin toss resulted in H (i.e., $\{HH, HT\}$ occurs), the updated table becomes:

Event	{HH}	{HT}	{TH}	{TT}
Probability				

Conditional Probability

- Without any additional information, the probability table is:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/4	1/4	1/4	1/4

- Given that the first coin toss resulted in H (i.e., $\{HH, HT\}$ occurs), the updated table becomes:

Event	{HH}	{HT}	{TH}	{TT}
Probability			0	0

Conditional Probability

- Without any additional information, the probability table is:

Event	$\{HH\}$	$\{HT\}$	$\{TH\}$	$\{TT\}$
Probability	$1/4$	$1/4$	$1/4$	$1/4$

- Given that the first coin toss resulted in H (i.e., $\{HH, HT\}$ occurs), the updated table becomes:

Event	$\{HH\}$	$\{HT\}$	$\{TH\}$	$\{TT\}$
Probability	$1/2$	$1/2$	0	0

Conditional Probability

- Without any additional information, the probability table is:

Event	$\{HH\}$	$\{HT\}$	$\{TH\}$	$\{TT\}$
Probability	$1/4$	$1/4$	$1/4$	$1/4$

- Given that the first coin toss resulted in H (i.e., $\{HH, HT\}$ occurs), the updated table becomes:

Event	$\{HH\}$	$\{HT\}$	$\{TH\}$	$\{TT\}$
Probability	$1/2$	$1/2$	0	0

- Note that the probability of $\{HH\}$ increases from $1/4$ to $1/2$. This illustrates that the event $\{HH\}$ is dependent on the occurrence of the event $\{HH, HT\}$.

Conditional Probability

- Let B be an event with $P(B) > 0$ and $Q_B : \mathcal{A} \rightarrow [0, 1]$ be the probability measure reflecting the occurrence of B . For any $A \in \mathcal{A}$:

$$\begin{aligned} Q_B(A) &= Q_B(A \cap B) + Q_B(A \cap B^c) \\ &= Q_B(A \cap B) \\ &= k(B)P(A \cap B) \quad \text{for some } k(\cdot). \end{aligned} \tag{12}$$

By substituting $A = S$, we obtain $k(B) = 1/P(B)$. Therefore,

$$Q_B(A) = \frac{P(A \cap B)}{P(B)}. \tag{13}$$

Conditional Probability

- Let B an event with $P(B) > 0$ and $Q_B : \mathcal{A} \rightarrow [0, 1]$ be the probability measure reflecting the occurrence of B . For any $A \in \mathcal{A}$:

$$\begin{aligned} Q_B(A) &= Q_B(A \cap B) + Q_B(A \cap B^c) \\ &= Q_B(A \cap B) \\ &= k(B)P(A \cap B) \quad \text{for some } k(\cdot). \end{aligned} \tag{14}$$

By substituting $A = S$, we obtain $k(B) = 1/P(B)$. Therefore,

$$Q_B(A) = \frac{P(A \cap B)}{P(B)}. \tag{15}$$

- **Conditional Probability**: Let A and B be two events with $P(B) > 0$. The *conditional probability* of A given B , denoted by $P(A | B)$, is defined as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}. \tag{16}$$

Conditional Probability

- ▶ How do we reflect the occurrence of multiple events?
- ▶ Suppose we are informed that the first coin toss resulted in H (i.e., the event $B_1 = \{HH, HT\}$ occurs), which updates the probability table to:

Event	$\{HH\}$	$\{HT\}$	$\{TH\}$	$\{TT\}$
Probability	$1/2$	$1/2$	0	0

- ▶ Subsequently, we are further informed that the second coin toss resulted in T (i.e., the event $B_2 = \{HT, TT\}$ occurs). Then, the updated table becomes:

Event	$\{HH\}$	$\{HT\}$	$\{TH\}$	$\{TT\}$
Probability	0	1	0	0

- ▶ How do we formulate this? There are two intuitive approaches to consider:

$$Q_{B_1}(A | B_2) \text{ (Sequential update)} \quad \text{and} \quad P(A | B_1 \cap B_2) \text{ (Joint update)}. \quad (17)$$

Conditional Probability

- ▶ The two approaches are mathematically equivalent:

$$\begin{aligned}Q_{B_1}(A \mid B_2) &= \frac{Q_{B_1}(A \cap B_2)}{Q_{B_1}(B_2)} \\&= \frac{P(A \cap B_1 \cap B_2)/P(B_1)}{P(B_1 \cap B_2)/P(B_1)} \\&= \frac{P(A \cap B_1 \cap B_2)}{P(B_1 \cap B_2)} \\&= P(A \mid B_1 \cap B_2).\end{aligned}\tag{18}$$

- ▶ In a similar manner, we can define the conditional probability given multiple events as follows:

$$P(A \mid B_1 \cap B_2 \cap \cdots \cap B_n) = \frac{P(A \cap B_1 \cap B_2 \cap \cdots \cap B_n)}{P(B_1 \cap B_2 \cap \cdots \cap B_n)}.\tag{19}$$

Bayes' Rule

- ▶ As in the coin example, we are often interested in updating probabilities to reflect new information.
- ▶ Suppose we have a *partition* of the sample space, $B_1, \dots, B_n$ with $P(B_i) > 0$ such that:

$$\cup_{i=1}^n B_i = S \quad \text{and} \quad B_i \cap B_j = \emptyset \quad \text{when } i \neq j, \quad (20)$$

e.g., $B_1 = \{HH\}$, $B_2 = \{HT\}$, $B_3 = \{TH\}$, and $B_4 = \{TT\}$.

- ▶ Given the occurrence of A , how do we update the probability table from

$$P(B_1), P(B_2), \dots, P(B_n) \quad \text{where} \quad \sum_{i=1}^n P(B_i) = 1 \quad (21)$$

to

$$P(B_1 | A), P(B_2 | A), \dots, P(B_n | A) \quad \text{where} \quad \sum_{i=1}^n P(B_i | A) = 1? \quad (22)$$

Bayes' Rule

- **Bayes' Rule:** Let A and $B_1, \dots, B_n$ be events where the B_i are pairwise disjoint, $\overline{\bigcup_{i=1}^n B_i} = S$, and $P(B_i) > 0$ for all i . Then,

$$P(B_j | A) = \frac{P(A | B_j)P(B_j)}{\sum_{i=1}^n P(A | B_i)P(B_i)}. \quad (23)$$

Sketch of Proof: By definition, we have $P(B_j | A) = \frac{P(A \cap B_j)}{P(A)}$. We then apply the Law of Total Probability:

$$P(A) = \sum_{i=1}^n P(A | B_i)P(B_i). \quad (24)$$

Bayes' Rule

- ▶ Suppose three plants, B_1 , B_2 , and B_3 , produce 50%, 30%, and 20% of a company's total output, respectively.
- ▶ From historical quality control data, the defect rates for the items produced by these plants are 2%, 3%, and 5%, respectively.
- ▶ An inspector randomly selects an item from the combined output and finds it to be defective.
- ▶ **Q:** What is the probability that the defective item was produced by each of the plants?

Bayes' Rule

- ▶ Let D denote the event that a randomly selected item is defective.

$$\begin{aligned}P(B_1 \cap D) &= P(D \mid B_1)P(B_1) = 0.02 \times 0.50 = 0.010 \\P(B_2 \cap D) &= P(D \mid B_2)P(B_2) = 0.03 \times 0.30 = 0.009 \\P(B_3 \cap D) &= P(D \mid B_3)P(B_3) = 0.05 \times 0.20 = 0.010.\end{aligned}\tag{25}$$

- ▶ The total probability of a defect is $P(D) = 0.010 + 0.009 + 0.010 = 0.029$.
- ▶ Applying Bayes' Rule, the posterior probabilities are:

$$\begin{aligned}P(B_1 \mid D) &= \frac{0.010}{0.029} \approx 0.345 \\P(B_2 \mid D) &= \frac{0.009}{0.029} \approx 0.310 \\P(B_3 \mid D) &= \frac{0.010}{0.029} \approx 0.345.\end{aligned}\tag{26}$$

Odds

- ▶ Odds: The odds of an event A are defined as

$$\frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}. \quad (27)$$

- ▶ For instance, if $P(A) = 2/3$, then $P(A) = 2P(A^c)$, so the odds are 2. If the odds equal α , it is common to say that the odds are “ α to 1” in favor of the event.

Odds

- ▶ Consider a hypothesis H that is true with probability $P(H)$ ², and suppose that new evidence E is introduced.
- ▶ Then, the odds after observing E are:

$$\underbrace{\frac{P(H | E)}{P(H^c | E)}}_{\text{Posterior odds}} = \underbrace{\frac{P(H)}{P(H^c)}}_{\text{Prior odds}} \cdot \underbrace{\frac{P(E | H)}{P(E | H^c)}}_{\text{Likelihood ratio}}. \quad (28)$$

- ▶ That is, the posterior odds of H are equal to the prior odds multiplied by the ratio of the conditional probability of the evidence given H is true to that given H is false.

²The probability here may be interpreted as a degree of belief.

Odds

- ▶ Suppose we have two unfair coins: Coin H , which has a 75% probability of landing on H , and Coin T , which has a 25% probability of landing on H .
- ▶ We randomly pick one of the two coins with equal probability, yielding the following prior probabilities:

Event	{Coin H }	{Coin T }
Probability	1/2	1/2

- ▶ Given that the toss result is H , we can update the odds of having chosen Coin H :

$$\underbrace{\frac{P(\text{Coin } H \mid H)}{P(\text{Coin } T \mid H)}}_{\text{Posterior odds}} = \underbrace{\frac{P(\text{Coin } H)}{P(\text{Coin } T)}}_{\text{Prior odds}} \cdot \underbrace{\frac{P(H \mid \text{Coin } H)}{P(H \mid \text{Coin } T)}}_{\text{Likelihood ratio}} = \frac{1/2}{1/2} \cdot \frac{3/4}{1/4} = 3. \quad (29)$$

- ▶ Since the posterior odds are 3 to 1, the updated probability that we chose Coin H becomes 3/4.

Event Independence

- ▶ Independence: Two events A and B are said to be *independent* if:

$$P(A \cap B) = P(A)P(B). \quad (30)$$

- ▶ If $P(B) > 0$, this definition is mathematically equivalent to:

$$P(A | B) = P(A). \quad (31)$$

That is, the occurrence of B does not change the probability of A .

Event Independence

- ▶ For example, suppose we toss a fair die twice, yielding the sample space:

$$S = \{(1, 1), \dots, (6, 6)\}. \quad (32)$$

- ▶ Let $A = \{(3, 1), \dots, (3, 6)\}$ be the event that the first toss is a 3, and let $B = \{(1, 6), \dots, (6, 1)\}$ be the event that the sum of the two tosses is 7.
- ▶ Since $P(A \cap B) = P(A)P(B) = (1/6)(1/6) = 1/36$, A and B are independent. Intuitively, any first toss leaves exactly one valid second toss, keeping the conditional probability constant.
- ▶ Now, let $C = \{(2, 6), \dots, (6, 2)\}$ be the event that the sum is 8. Since $P(A \cap C) = P(\{(3, 5)\}) = 1/36 \neq P(A)P(C) = 5/216$, A and C are dependent. The first toss restricts the possibilities (e.g., 1 makes a sum of 8 impossible).

References I

- Deisenroth, M. P., Faisal, A. A., and Ong, C. S. (2020). *Mathematics for machine learning*. Cambridge University Press.
- Rice, J. A. (2007). *Mathematical statistics and data analysis*, volume 371. Thomson/Brooks/Cole Belmont, CA.
- Ross, S. M. (2020). *A first course in probability*. Pearson Harlow, UK.